How much is the wind worth? Wind effects on sprint and horizontal-jump performance across age and ability in a national competition registry

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Running head: Wind effects across age and ability

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Abstract

Almost all wind-effect estimates in athletics come from elite seniors, yet the ±2.0 m/s legality rule applies equally to ten-year-olds. Using the Norwegian Athletics Federation's registry, we estimated wind effects within athlete-seasons in five sprints and four horizontal jumps at ages 10–34 (392 797 results). Per m/s of wind (linear fit over head- and tailwinds), results changed by 0.059 s on 60 m, 0.082 s on 100 m, 0.121 s on 200 m, 3.2 cm in the long jump and 4.2 cm in the triple jump. Absolute sprint effects declined with age (200 m, males: 0.144 s per m/s at 10–11 to 0.085 at 20–34) and ability, scaling approximately with race duration. Headwinds cost more than tailwinds gave (100 m: 0.107 vs. 0.065 s per m/s), on the 100 m by about as much as drag physics predicts at athletes' own speeds, in both sexes. A legal +2.0 m/s tailwind was worth 0.11–0.25 s on the 100 and 200 m from age 12, roughly equal across ages for boys, larger for girls than for senior women, and enough to reorder 2025 season lists at youth and senior ages. Age-specific corrections can make lists, records and talent evaluation fairer.

Keywords: wind; sprint; long jump; youth athletics; age-group competition; registry study

1 Introduction

Wind is the one weather variable athletics regulates. Sprint and horizontal-jump results count for records only when tailwind does not exceed 2.0 m/s, gauge readings are part of the official result, and seasonal lists in many countries separate legal from wind-aided marks. How much the wind actually moves a result has been estimated in two ways: physical and quasi-physical models of the sprint (Linthorne, 1994; Mureika, 2001), and data-driven analyses of elite competition results (Moinat et al., 2018; Hollings et al., 2012). Both traditions agree on the order of magnitude for elite seniors — roughly 0.04–0.06 s per m/s in the 100 m, somewhat more for women than for men — and both concern elite seniors only. The one population study of non-elite sprinters we know of, on US high-school 100 m results, found that headwinds slowed girls and boys more than a drag model for elite men predicts and attributed the excess to psychology (Lybbert and Zheng, 2023); it did not examine how the effects scale with age or ability. Reviews of sprint testing likewise treat wind as a condition to standardise, with correction factors derived from adult data (Haugen and Buchheit, 2016).

Yet nothing guarantees that elite estimates transfer down the age and ability range, and the physics pulls in two directions. A slower runner spends longer on the course — a 12-year-old needs roughly a third more time over 100 m than a senior — so any given wind acts on them longer, inflating the absolute time effect. But aerodynamic drag scales with the square of airspeed, so slower runners spend a smaller share of their power against air, deflating it; children's higher surface-to-mass ratio pushes the other way again. The net scaling of the wind effect across age and ability is an empirical question that elite data cannot answer, and it matters practically: age-group records, seasonal youth lists, age-graded scoring tables and qualification standards all treat marks within ± 2.0 m/s as equivalent, in every age class, and talent evaluation leans on exactly these marks (Guttormsen, 2026).

The Norwegian registry is unusually well placed to answer it. Every sanctioned result in Norway is registered, and wind is measured and recorded down to the outdoor 60 m sprints that children contest from age 10 — to our knowledge the only population-scale source of wind-measured sprint data on children. We pre-specified five questions: the direction of the effect within athlete and season (H1, a design check); how the absolute effect scales with age (H2

— the exposure-time and aerodynamic-share mechanisms predict opposite directions); how the percentage effect scales (H3); whether headwinds cost more than tailwinds give (H4, as aerodynamics predicts); and what the ± 2.0 m/s window is worth in each class, expressed against athletes' own performance variability (H5). The analysis plan was committed to version control before estimation; the study was not externally preregistered. After the main estimations, by a post-plan decision, class-specific short hurdles were added as a documented extension.

2 Methods

2.1 Data

We used a frozen extract (29 October 2026) of the Norwegian Athletics Federation's results database, which holds results electronically from 2011 (with thinner coverage in 2011, the first electronic season). The main analysis covers all results from 2011 onwards in nine wind-measured events: 60 m, 80 m, 100 m, 150 m and 200 m sprints, and long jump, long jump from a take-off zone, triple jump and triple jump from a zone (in the zone variants, used in the youngest classes, take-off is from a marked zone rather than a board, which reduces fouls). The hurdles extension covers the ten short-hurdle event codes with substantive wind data (60 m to 110 m hurdles; each code specifies the class's hurdle height, so a code identifies an age-class specification). Only approved, outdoor, electronically timed results of athletes with known sex and birth year were analysed, at calendar-year ages 10–34, with event-specific plausibility bounds and wind readings within ± 8 m/s. Before these filters we removed duplicate records: 47 962 board-jump rows that re-imports in 2026 had copied from an identical take-off-zone result (same athlete, date, mark and wind; almost all at ages 11–13, and listed in the source only under the zone event), 8 907 such copies from multi-day meets, dated to the meet's first day while the zone result lies one to six days later, and 770 exact duplicates, mostly meets stored twice. Because the 2026 import filed take-off-zone jumps under the board events, the 1 597 board-event jumps at ages 11–13 in 2026 were also excluded. Of 670 803 extracted rows in the main events, 392 797 entered the analysis. The largest exclusions were indoor results — principally the indoor 60 m (86 449 rows), 200 m (35 849) and long jump (25 862) — and

16 246 outdoor rows without a wind value (overall, 4.8 % of approved outdoor results lack a reading). Missing wind is a completeness rather than a selection concern: it is concentrated in the smallest meets (9.0 % of rows in the smallest third of meets by size, against 1.4 % in the largest third) and its yearly share fluctuates between 2 and 7 %. Of the analysed rows, 295 293 lie in athlete-season-event cells with at least two wind-measured results and identify the estimates; the hurdles extension contributed 33 205 analysed rows. Times are analysed in seconds and jumps in metres (reported in centimetres); units were verified against known magnitudes per event.

The same estimator was then applied, after the main analyses, to two external samples of wind-measured outdoor results in the 100 m, 200 m, 100 and 110 m hurdles, long jump and triple jump: all such results in the TFRRS profiles of NCAA Division I athletes, 2008–2026, read from a cleaned, hash-checked shared build of the profiles (546 854 results of 36 540 athletes after the registry's filters, with combined-event sub-results excluded), and the career results, as listed by Tilastopaja, of athletes who reached a World Championships, Olympic or other major championship final between 1983 and 2026 (66 589 results of 2 219 athletes). Neither source gives every athlete's age, so they are compared with the registry at their mean race times, the collegiate sample also by ability tertile.

2.2 Statistical model

The identifying variation is within athlete × season × event: the same athlete, in the same event and season, racing under different winds. For each event we estimated

$$y_{it} = \alpha_{i(\text{athlete}\times\text{year})} + \beta \cdot \text{wind}_{it} + \text{month dummies} + \text{round dummies} + \varepsilon_{it},$$

by within-demeaning; 74 110 athlete-season-event cells contribute at least two wind-measured results, and estimates are reported only where at least 300 results in 100 athlete-seasons contribute. Because the gauge reading is constant within a race, the race — and conservatively the meet — is the natural clustering level (Moulton, 1986); standard errors are therefore clustered on meet, counting the absorbed athlete-season effects in the small-sample correction because

they are not nested within meets, with athlete clustering (the analysis plan's original choice, which gives intervals roughly half as wide in the large sprints and somewhat narrower ones elsewhere) retained as robustness. The month dummies are pooled across years and capture a common within-season profile; year-specific shocks are absorbed only in the meet fixed-effects specification. β is reported per m/s of wind, signed as a gain with tailwind, in seconds or centimetres, and in percent from a log-outcome specification; because headwinds cost more than tailwinds give, this linear slope averages the two sides, and statements about tailwinds use the separate tailwind slope. Age-band (10–11 through 20–34) and sex-specific models address H2/H3; ability tertiles address ability scaling, with ability defined as the athlete-season mean of *wind-adjusted* performance (each mark corrected to still air with the event-level slope), so that the ability measure itself is not contaminated by wind exposure, and the contributing athlete-seasons divided into equal thirds within event $\times$ sex $\times$ age band; separate slopes for head- and tailwind address H4; a quadratic term probes curvature. For H5, the advantage of a +2.0 m/s tailwind over still air is twice the slope over 0 to +2 m/s, estimated for each event, sex and age band in a piecewise model with separate slopes for headwinds and for tailwinds above +2 (the response flattens above +2, so a single tailwind slope would understate the window, and the symmetric slope would overstate it). It is expressed in seconds and as a share of a robust within-athlete-season scale: 1.4826 times the median absolute deviation of within-cell residuals of wind-adjusted marks, over cells with at least three results and corrected for the estimated cell means. The robust scale is needed because the event-wide plausibility bounds admit a few gross recording errors (for example 200 m times filed as 100 m), which barely move the slopes but inflate a classical standard deviation. Pre-specified robustness analyses required at least three results per cell, excluded readings of exactly 0.0, split the period (2011–2019 vs. 2021–2026; 2020, whose season was curtailed by the pandemic, belongs to neither half), and replaced month dummies with meet fixed effects (two-way demeaning); a post-plan check excluded readings within $\pm$0.2 of +2.0. Other departures from the plan: H5 is valued from the 0 to +2 m/s slope against a robust scale and for the sprints only (the plan named the +2.0 m/s advantage against the classical within-cell standard deviation, for all events); ability is the wind-adjusted rather than the raw season mean; the planned formal test of the age gradient (H2) was not run, and the

age bands are reported with their intervals instead; the linear model uses readings within ±8 rather than ±5 m/s; and the drag-only benchmark, the per-second scaling, meet fixed effects by age band, the age-graded points, the 2025 lists, the sex- and age-specific asymmetry comparison (Table S7) and the external-sample comparison (Table S8) were added after the plan. Analyses used Python 3.13.7 with pandas 2.3.3, numpy 2.4.1 and statsmodels 0.14.6.

Two interpretive rules were fixed in advance. First, a single gauge reading does not measure the wind each athlete met; to the extent this error is classical it attenuates estimates toward zero, so the sprint effects are read as lower bounds. The measurement protocol also differs by event family: under World Athletics rules the gauge beside the straight averages 10 s from the gun in the 100 m (13 s in the short hurdles), 10 s from when the leader enters the straight in the 200 m — so it measures the wind over only part of a course that is half bend — and 5 s per attempt in the jumps; Norwegian practice applies the same conventions to the domestic distances (60 and 80 m as the 100 m; 150 m as the 200 m). Comparisons across event families bear this in mind. Second, wind is not randomly assigned: identification relies on within-season comparisons with month, round and (in robustness) meet controls, and no causal language beyond this design sense is used. For the jumps, one further caveat applies: the registry stores only the wind of the measured (best) attempt, which describes the conditions of a series imperfectly, and the direction of the resulting bias is not established; the lower-bound reading is therefore applied to the sprints only.

2.3 Sample size estimation and justification

No sample size was calculated in advance, because the study used every result in the registry that met the inclusion criteria (Section 2.1): the sample is the eligible population rather than a sample drawn from it. Precision was instead governed by two rules: estimates are reported only where at least 300 results in 100 athlete-seasons contribute, and standard errors are clustered on meet, which gives wider intervals than the athlete clustering of the analysis plan. At the event level the resulting 95 % confidence intervals have half-widths of 0.002–0.009 s per m/s in the 60, 100 and 200 m (0.013 and 0.037 in the small 80 and 150 m events) and 0.2–1.0 cm in the four jump events; age-band estimates that rest on thin samples are marked in the tables.

2.4 Ethics and transparency

The data are competition results registered by the Norwegian Athletics Federation: results from age 13 are publicly listed, while records for the 10–12 age classes are registered but not publicly released by the federation. No athlete is identified. Performance data below age 13 are used solely to estimate measurement conditions — not to rank, list or evaluate children — in keeping with the purpose of the Norwegian children's sport regulations; no result of an athlete under 13 appears in ranked, listed or identifiable form in the paper or its outputs. The study is not medical or health research as defined by Norway's Health Research Act (Helseforskningsloven §§ 2 and 4), so approval by a Regional Committee for Medical and Health Research Ethics was not required, and Sikt (the Norwegian Agency for Shared Services in Education and Research) confirmed that the use of the data raises no data-protection concerns. No participants were contacted or recruited, and informed consent was not required. The analysis plan was committed to version control before estimation; athlete-level files are regenerable from the registry. Reporting follows STROBE where applicable (von Elm et al., 2007).

3 Results

3.1 Sample and wind exposure

Table 1 describes the nine main events. Wind readings centred near calm with wide spread (median +0.3 m/s; 5th–95th percentile −2.1 to +2.7). The distribution shows digit preference — 8.7 % of readings are exactly 0.0, more than any continuous wind distribution would place there and consistent with some unoperated gauges being recorded as calm, and 16 % fall on whole metres per second — and thins just above the legality line: 3.4 % of readings lie in (1.8, 2.0] against 2.2 % in (2.0, 2.2], with 1.6 % exactly +2.0. Both patterns motivate robustness checks: excluding exact zeros, and excluding readings within ±0.2 of +2.0, leave every estimate essentially unchanged (Table S2). The bulk of the material is young: the 60 m, contested outdoors by the youngest classes, is the largest single event (103 286 analysed results, of which 72 558 lie in athlete-seasons with at least two wind-measured races and thus contribute to estimation).

Table 1: Analysed events: volume, marks, and wind exposure (2011–2026).

Event	Results	In cells ≥2	Athlete-seasons	Athletes	Median mark	Wind, median (p5–p95)
60 m	103 286	72 558	19 354	10 790	9.59 s	+0.1 (−2.3–+2.5)
80 m	5 795	1 410	620	466	12.53 s	+0.0 (−2.6–+2.5)
100 m	76 430	64 557	13 222	5 451	12.88 s	+0.4 (−2.2–+2.8)
150 m	2 151	328	150	125	19.64 s	+0.3 (−2.5–+2.2)
200 m	76 113	59 246	15 549	6 608	27.99 s	+0.4 (−2.2–+2.9)
Long jump	51 866	35 774	8 126	4 365	4.42 m	+0.5 (−1.6–+2.7)
Long jump (zone)	54 665	44 336	12 338	7 835	3.64 m	+0.3 (−1.7–+2.7)
Triple jump	14 631	11 658	2 890	1 543	10.31 m	+0.7 (−1.3–+2.9)
Triple jump (zone)	7 860	5 426	1 861	1 413	8.42 m	+0.4 (−1.5–+2.8)

Results: analysed rows (approved, outdoor, electronic, ages 10–34, wind within ±8 m/s). In cells ≥2: the estimation sample (the n of Table 2). Athlete-seasons and athletes: units contributing to estimation in that event. Counts and wind summaries are computed from the analysis file; the estimation files are available on request.

3.2 What a metre per second is worth (H1)

Table 2 reports the main within-athlete estimates. Per m/s of wind, results changed by 0.059 s (95 % CI 0.057–0.061) on 60 m, 0.082 s (0.080–0.085) on 100 m, 0.121 s (0.113–0.130) on 200 m, 3.2 cm (2.9–3.5) in the long jump and 4.2 cm (3.5–4.8) in the triple jump; in percentage terms, 0.39–0.68 % per m/s. All directions matched H1. The small 80 m and 150 m events gave consistent estimates with wide intervals.

Table 2: Effect per m/s of wind (linear fit over head- and tailwinds, signed as a gain with tailwind), within athlete × season (95 % CIs clustered on meet).

Event	Per m/s, linear (95 % CI)	Percent per m/s (95 % CI)	n	Athlete-seasons
60 m	0.059 s (0.057–0.061)	0.626 (0.602–0.649)	72 558	19 354
80 m	0.075 s (0.062–0.088)	0.630 (0.531–0.729)	1 410	620
100 m	0.082 s (0.080–0.085)	0.646 (0.625–0.666)	64 557	13 222
150 m	0.081 s (0.045–0.118)	0.467 (0.278–0.656)	328	150
200 m	0.121 s (0.113–0.130)	0.442 (0.410–0.474)	59 246	15 549
Long jump	3.16 cm (2.86–3.46)	0.676 (0.604–0.747)	35 774	8 126
Long jump (zone)	2.11 cm (1.90–2.32)	0.570 (0.513–0.628)	44 336	12 338
Triple jump	4.18 cm (3.53–4.84)	0.394 (0.331–0.458)	11 658	2 890
Triple jump (zone)	3.88 cm (2.87–4.89)	0.449 (0.331–0.568)	5 426	1 861

Point estimates are identical under athlete clustering (the analysis plan's original choice), whose intervals are roughly half as wide in the large sprints and somewhat narrower elsewhere; both sets are in the estimation files available on request.

The published elite figures of roughly 0.04–0.06 s per m/s on the 100 m (Linthorne, 1994; Moinat et al., 2018) mainly describe tailwinds. On the tailwind side alone, the population slopes were 0.047 s (60 m), 0.065 s (100 m) and 0.078 s (200 m) per m/s (Table 4), so the 100 m figure is about 1.1–1.6 times the elite range. Restricted to seniors (20–34), the 100 m tailwind slope was 0.064 s for men and 0.057 s for women, at or just above the elite range — and our senior men, averaging about 11.4 s on the 100 m, are themselves slower than the elite samples. The elite literature describes the fast end of a gradient. Applying the estimator unchanged to the two external samples places that end on the registry's scale (Table S8): on the 100 m, 0.054 s per m/s (95 % CI 0.053–0.056) for collegiate men and 0.058 (0.056–0.060) for collegiate women, 0.048 (0.045–0.051) and 0.059 (0.055–0.063) for championship finalists — 58–72 % of the registry's 0.082, and for three of the four groups within 0.002 s of the level formula of Moinat et al. (2018) at their mean times.

3.3 The effect falls with age — and with speed (H2, H3)

Figure 1 and Table 3 show the core result. In absolute seconds, the wind effect declined with age in every sprint and both sexes — monotonically in every series except the sparse girls' 200 m at 10–11: on 200 m for boys and men from 0.144 s per m/s at 10–11 to 0.085 at 20–34; on 60 m from 0.070 at 10–11 to 0.041 at 16–17; on 100 m from 0.101 at 12–13 to 0.071 at 20–34. Exposure time accounts for most of the gradient: mean race durations shrink in almost the same proportion (men's 100 m, 14.97 s at 12–13 vs. 11.42 s at 20–34, a ratio of 1.31, against a wind-effect ratio of 1.42), so the percentage effect (Figure 2) is far flatter — but not constant, declining for instance from 0.70 to 0.62 % per m/s on the men's 100 m and from 0.70 to 0.52 % on the 60 m. Dividing each event's effect by its mean race duration makes the scaling explicit (Table S4): the straight sprints line up at 0.0059–0.0063 s per m/s per second of race time, while the bend events sit a third lower (150 m 0.0040, 200 m 0.0043) — as expected when the gauge, read beside the home straight, measures the wind over only part of a course that is half bend, and when the wind meets the runner at varying angles through the bend. In the jumps the absolute effect for girls and women ran the other way, rising with age in centimetres (long jump: 2.9 cm per m/s at 14–15 to 4.1 at 20–34) as jump distances and flight times grow, but not for boys and men (3.4 at 14–15, 3.1 at 20–34), and the percentage effect shows no consistent age direction (Table S1); below 14 nearly all horizontal jumps are made from a take-off zone, so the board-jump series start at 14–15. Within age bands, ability scaling follows the sprint pattern (Figure 3): with ability measured as the wind-adjusted season mean, the lowest tertile gained most per m/s in every sprint (100 m: 0.088 s against 0.079 in the highest tertile; 60 m: 0.066 against 0.056; 200 m: 0.133 against 0.117), while the jump tertiles did not separate. Slower athletes — whether young or simply weaker — are exposed longer, and no single correction factor, absolute or proportional, is exactly age-invariant. Within the collegiate sample the same proportionality holds (its ability tertiles sit at 0.0047–0.0051 per second of race time on the 100 m), but across samples it does not: per second of race time the 100 m rate is 0.0062–0.0064 in the registry against 0.0047–0.0052 in the external samples, 15–27 % lower for the same sex, so the law of Table S4 predicts 0.061–0.077 s per m/s at their mean times against the observed 0.048–0.059 (Table S8). The gap persists at equal race times: the registry's seniors gained 0.071

s per m/s (0.066–0.075) against 0.057 (0.053–0.061) for the slowest collegiate tertile of men, at 11.4 and 11.3 s, and 0.072 (0.067–0.078) against 0.060 (0.057–0.064) for women, at 12.6 and 12.8 s. The per-second rate thus carries a sample-specific component — gauge practice, weather or population — that comparisons with elite estimates must allow for.

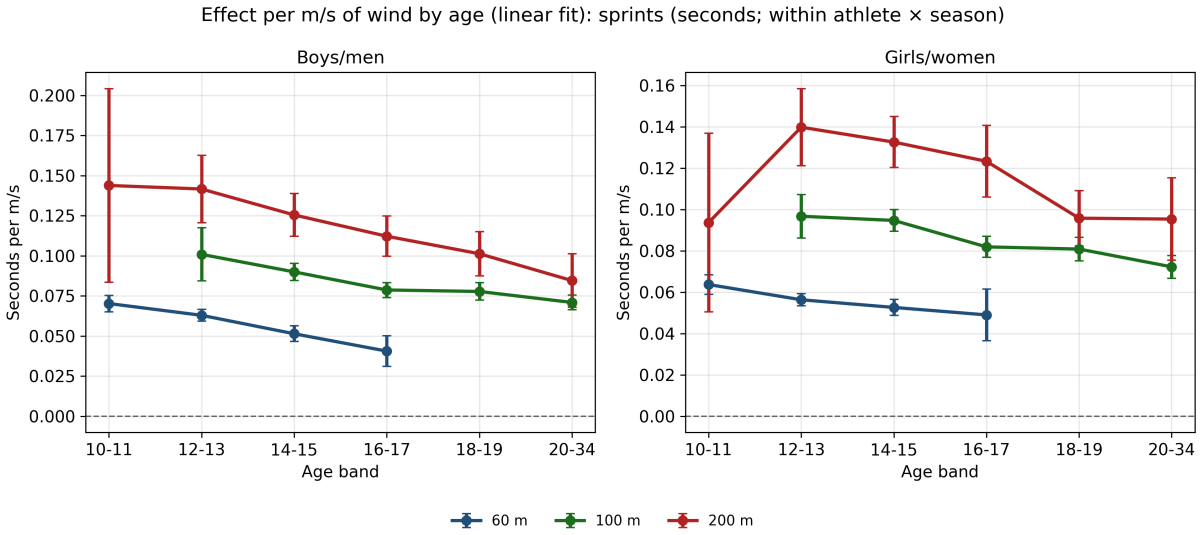


Figure 1: Effect per m/s of wind (linear fit over head- and tailwinds) by age band, in seconds, for sprints (within athlete × season; 95 % CIs clustered on meet).

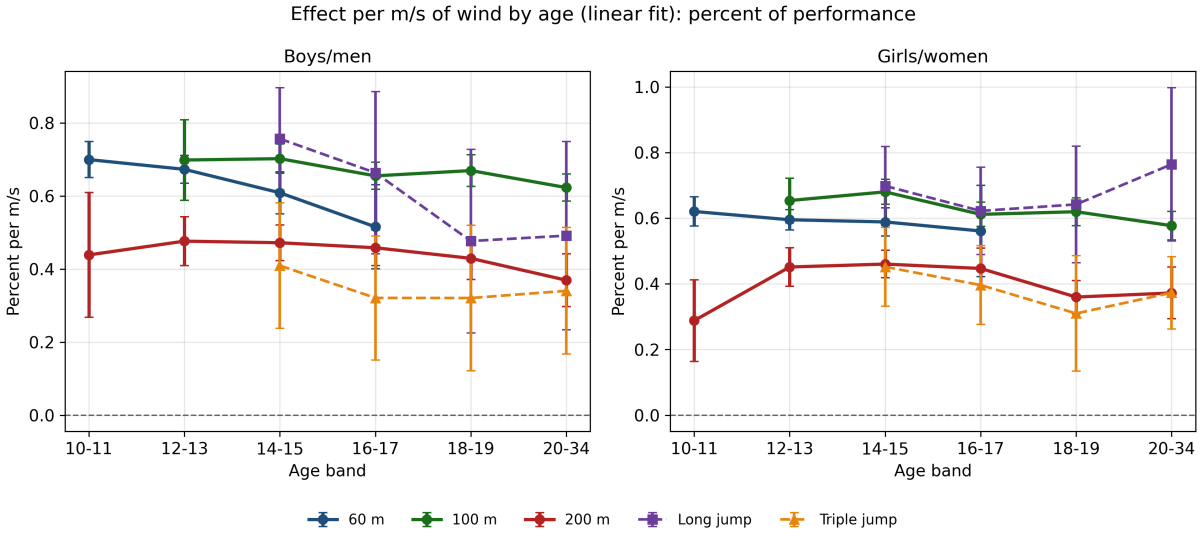


Figure 2: The same scaling in percent of performance, for sprints (solid lines) and horizontal jumps (dashed). For the sprints the percentage effect is far flatter across age than the absolute effect, but still declines; the jumps show no consistent age direction (95 % CIs clustered on meet).

Table 3: Effect per m/s of wind (linear fit; seconds, 95 % CI clustered on meet) by age band and sex, sprints.

Event	Sex	10–11	12–13	14–15	16–17	18–19	20–34
60 m	Boys/men	0.070 (0.065–0.075)	0.063 (0.059–0.067)	0.051 (0.046–0.056)	0.041 (0.031–0.050)	—	—
60 m	Girls/women	0.064 (0.059–0.068)	0.056 (0.053–0.059)	0.053 (0.049–0.057)	0.049 (0.037–0.061)	—	—
100 m	Boys/men	—	0.101 (0.084–0.117)	0.090 (0.085–0.095)	0.079 (0.074–0.083)	0.078 (0.072–0.083)	0.071 (0.066–0.075)
100 m	Girls/women	—	0.097 (0.086–0.107)	0.095 (0.089–0.100)	0.082 (0.077–0.087)	0.081 (0.075–0.087)	0.072 (0.067–0.078)
200 m	Boys/men	0.144 (0.083–0.204)	0.142 (0.121–0.163)	0.125 (0.112–0.139)	0.112 (0.100–0.125)	0.101 (0.088–0.115)	0.085 (0.068–0.101)
200 m	Girls/women	0.094 (0.050–0.137)	0.140 (0.121–0.159)	0.133 (0.120–0.145)	0.123 (0.106–0.141)	0.096 (0.082–0.109)	0.095 (0.075–0.115)

Dashes: fewer than 300 results in 100 athlete-seasons, the reporting threshold (60 m above 17, which is rarely run outdoors; 100 m at 10–11, where 60 m is the standard sprint). Table values are reproduced from the estimation files available on request.

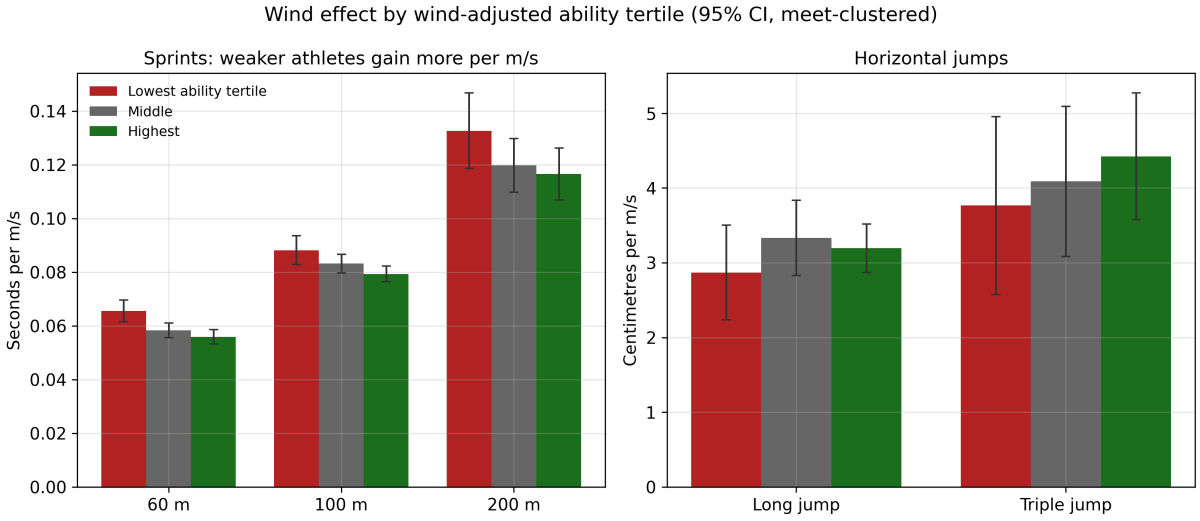


Figure 3: Wind effect by ability tertile: athlete-seasons divided into equal thirds of the wind-adjusted season mean within event × sex × age band (95 % CIs clustered on meet).

3.4 Headwinds cost more than tailwinds give (H4)

In every large sprint event the headwind slope exceeded the tailwind slope (Figure 4; Table 4): on 100 m a -1 m/s headwind cost 0.107 s against a +1 m/s tailwind's 0.065 s ($p < 0.001$ for equal slopes); on 200 m, 0.184 against 0.078; on 60 m, 0.072 against 0.047. The positive quadratic terms tell the same story (200 m: +0.0141 s per $(\text{m/s})^2$, SE 0.0013; Table S5). Drag proportional to the square of airspeed predicts this direction, and on the 100 m also roughly this size. The familiar ratio $(2v+1)/(2v-1) \approx 1.14$–$1.18$ describes a wind of exactly ± 1 m/s, not

slopes averaged over the observed range of winds; imposing drag-only physics (time change proportional to the change in squared airspeed at the athlete's mean speed, scaled so that its linear slope equals the observed one; only the head/tail ratio and the curvature relative to the slope are therefore predictions, and the mean speed, which includes the start, overstates the predicted asymmetry most on the 60 m) on the observed winds and fitting the same piecewise model predicts head/tail ratios of 1.84 (60 m), 1.72 (100 m) and 1.85 (200 m), against observed 1.52, 1.66 and 2.36, and quadratic terms of 0.0048, 0.0063 and 0.0099 against observed 0.0037, 0.0057 and 0.0141. The 100 m thus matches drag closely, the 60 m is somewhat less asymmetric, and only the 200 m — where the gauge reads only the home straight — is more asymmetric than drag alone predicts. On the 100 m the asymmetry was larger for girls and women than for boys and men (head/tail ratio 1.77, 95 % CI 1.58–1.96, against 1.54, 1.36–1.71), as drag predicts for slower runners (1.73 and 1.67 at their own speeds); in 9 of the 10 sex $\times$ age bands the drag-only ratio lay inside the observed interval, and the exception lay below it (Table S7). Within meets, where weather is held fixed, the ratios were 1.59 and 1.46 against drag-only 1.70 and 1.63. The collegiate 100 m ratios likewise matched drag at the athletes' own speeds (1.62 and 1.61 against 1.67 and 1.76, men and women), whereas the championship finalists exceeded it (1.93 and 1.86 against 1.45 and 1.49; Table S8); and in all three samples the tailwind slope above +2.0 m/s was 12–33 % lower than over 0 to +2 m/s, against the 26–36 % that drag-only physics predicts for the observed winds, so the legality line leaves no mark on the response beyond physics. In the jumps the two slopes did not differ significantly in the long jump (2.93 against 3.71 cm, $p = 0.18$), and in the zone variants the tailwind slope was the steeper one; since the registry stores only the wind of the measured attempt, the jump asymmetries are not interpreted.

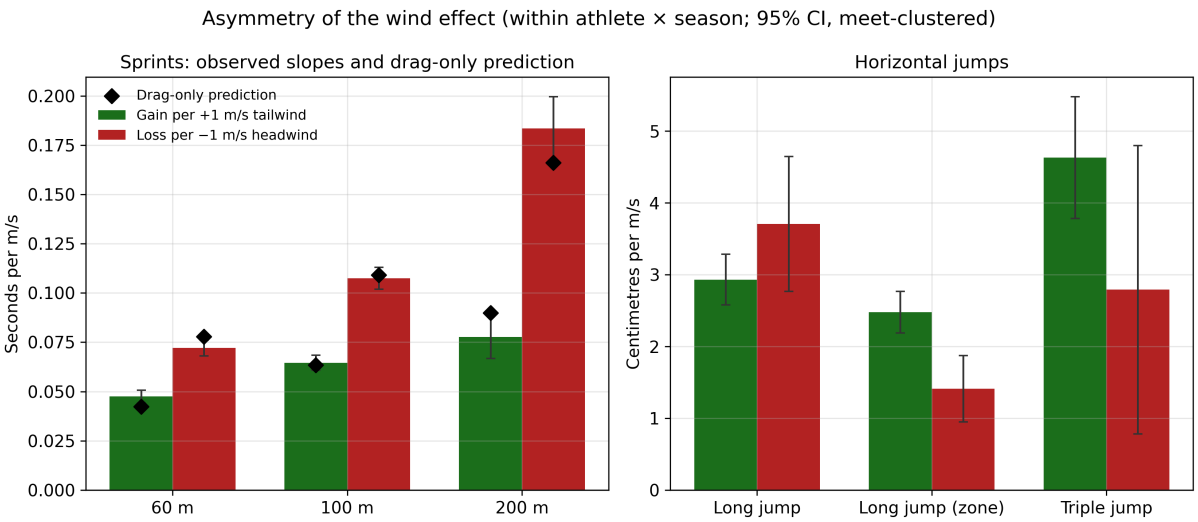


Figure 4: Separate head- and tailwind slopes (within athlete × season; 95 % CIs clustered on meet). Black diamonds: the slopes that drag-only physics predicts for the observed winds (sprints), with the drag response scaled to the observed linear slope.

Table 4: Asymmetry: separate tail- and headwind slopes, observed head/tail ratio, and the ratio drag-only physics predicts for the same estimator (95 % CIs clustered on meet).

Event	Unit	Tailwind gain per +1 m/s	Headwind loss per −1 m/s	Ratio	Drag-only ratio	p equal slopes
60 m	s	0.047 (0.044–0.051)	0.072 (0.068–0.076)	1.52	1.84	<0.001
80 m	s	0.072 (0.048–0.097)	0.078 (0.054–0.101)	1.07	1.72	0.805
100 m	s	0.065 (0.061–0.068)	0.107 (0.102–0.113)	1.66	1.72	<0.001
150 m	s	0.093 (0.031–0.154)	0.067 (−0.031–0.164)	0.72	1.80	0.721
200 m	s	0.078 (0.067–0.089)	0.184 (0.168–0.199)	2.36	1.85	<0.001
Long jump	cm	2.93 (2.58–3.28)	3.71 (2.77–4.65)	1.27	—	0.177
Long jump (zone)	cm	2.48 (2.19–2.77)	1.41 (0.95–1.87)	0.57	—	<0.001
Triple jump	cm	4.63 (3.78–5.48)	2.79 (0.78–4.80)	0.60	—	0.136
Triple jump (zone)	cm	4.69 (3.40–5.98)	1.90 (−0.25–4.04)	0.40	—	0.044

Drag-only ratio: time changes proportional to the change in squared airspeed at the athlete's mean speed, scaled to the observed linear slope, imposed on the observed winds and estimated with the same piecewise model. The simpler ratio $(2v + 1)/(2v - 1)$ for a wind of exactly ± 1 m/s is 1.14–1.18. No benchmark is given for the jumps, where wind acts on run-up and flight differently.

3.5 Short hurdles: like flat sprints over 60 m, mostly larger over 100 and 110 m

In the age-class system, hurdle events carry class-specific heights, so each event code identifies an age-class specification. Across the ten codes with wind data (Table 5; Figure 5), the 60 m hurdles matched the flat 60 m of the same sex and class: 0.055–0.067 s per m/s against 0.056–0.070, point estimates from 21 % below to 9 % above the flat references, with overlapping intervals. Over 100 and 110 m the hurdle estimates mostly exceeded the flat 100 m reference, from 4 % below to 48 % above; in the girls' 76.2 cm class (0.108 against 0.082) the hurdle interval lies entirely above the flat interval, and in the boys' 84 cm class (0.133 against 0.090) it lies above the flat point estimate but overlaps the flat interval. Per second of race time the pooled hurdle codes (0.0048–0.0069 s per m/s) span the straight-sprint rates (Table S4), so the hurdles' longer race times account for much of the difference, though not all of it in those two youth classes. Within the 100 m hurdles the youth codes also exceeded the senior code (0.108 s per m/s for girls' 76.2 cm at 16–17 and 0.133 for boys' 84 cm at 14–15, against 0.069 for the senior women's 84 cm event), although these comparisons cross sexes and hurdle heights.

Table 5: Wind effect in short hurdles by class-specific event code (sex-specific rows; flat-sprint reference for the same sex and distance class in that sex's modal age band for the code; 95 % CIs clustered on meet).

Event (hurdle height)	Sex	Modal band	Gain per +1 m/s (95 % CI), s	Flat reference (95 % CI), s	n
60 m hurdles 68 cm	Boys	10–11	0.055 (0.033–0.077)	0.070 (0.065–0.075)	1 214
60 m hurdles 68 cm	Girls	10–11	0.067 (0.048–0.086)	0.064 (0.059–0.068)	2 562
60 m hurdles 76.2 cm	Boys	12–13	0.063 (0.048–0.078)	0.063 (0.059–0.067)	3 435
60 m hurdles 76.2 cm	Girls	12–13	0.062 (0.051–0.073)	0.056 (0.053–0.059)	7 222
80 m hurdles 76.2 cm	Girls	14–15	0.072 (0.051–0.092)	—	2 520
80 m hurdles 84 cm	Boys	14–15	0.078 (0.051–0.104)	—	1 473
100 m hurdles 76.2 cm	Girls	16–17	0.108 (0.089–0.128)	0.082 (0.077–0.087)	1 046
100 m hurdles 84 cm	Boys	14–15	0.133 (0.092–0.174)	0.090 (0.085–0.095)	709
100 m hurdles 84 cm	Women	20–34	0.069 (0.041–0.097)	0.072 (0.067–0.078)	2 842
100 m hurdles 91.4 cm	Boys	16–17	0.084 (0.052–0.115)	0.079 (0.074–0.083)	679
110 m hurdles 91.4 cm	Boys	16–17	0.103 (0.035–0.171)	0.079 (0.074–0.083)	583
110 m hurdles 100 cm	Men	18–19	0.096 (0.036–0.155)	0.078 (0.072–0.083)	656
110 m hurdles 106.7 cm	Men	20–34	0.084 (0.047–0.121)	0.071 (0.066–0.075)	588

The hurdles analysis was added after the main estimations by a post-plan decision and is documented as a post-plan extension. Flat references are the 60 m or 100 m estimate for the same sex in that sex's modal age band; the flat 80 m has too few results at 14–15 for a reference. Codes with fewer than 300 results for one sex have only the other sex's row.

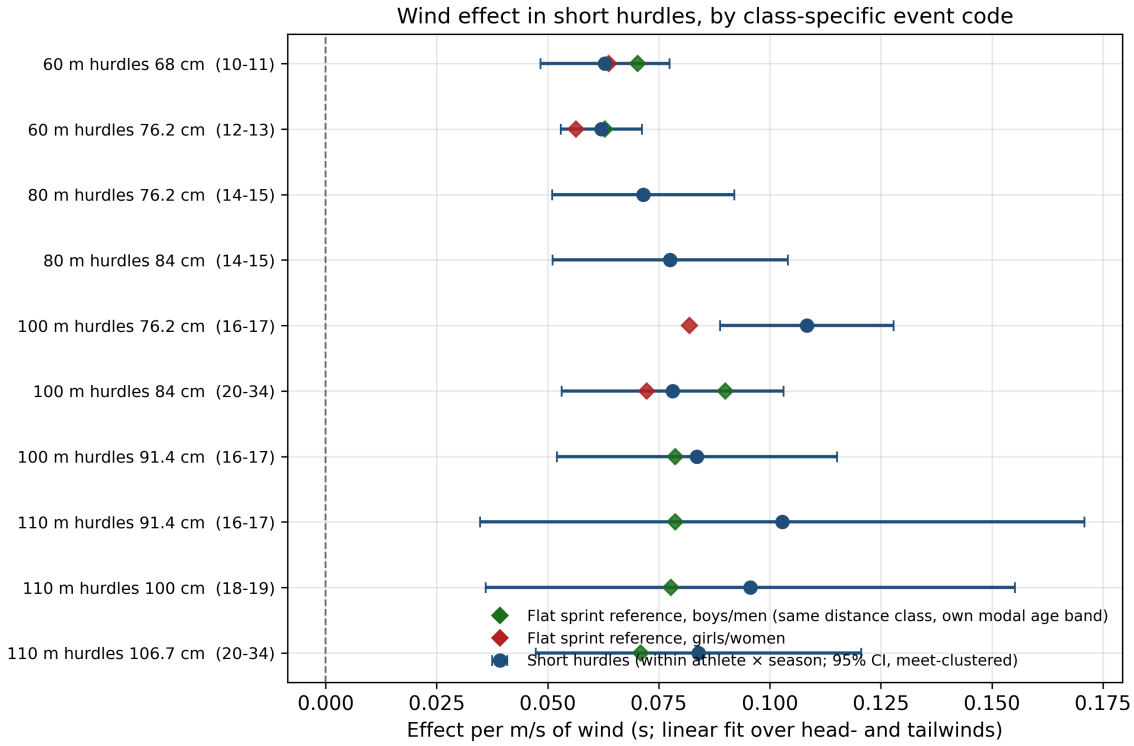


Figure 5: Short hurdles: wind effect per class-specific code (pooled sexes; dots with 95 % CIs clustered on meet) against flat-sprint references for each sex (diamonds; boys/men green, girls/women red).

3.6 What the legal window is worth in each class (H5)

Table 6 translates the estimates into the quantity the rules regulate: the advantage of a +2.0 m/s tailwind over still air, valued from the slope over 0 to +2 m/s (Section 2.2) and expressed in seconds and as a share of a robust measure of the athlete's own within-season variability. From age 12 the window was worth 0.11–0.25 s on the 100 and 200 m. For boys and men it showed no clear age trend in seconds: 0.14–0.16 s on the 100 m and 0.13–0.19 s on the 200 m in the bands from 12–13 to 20–34, with overlapping intervals. On the boys' 100 m the age decline of the linear slope (Section 3.3) lies mainly in the headwind slope (Table S3). For girls and women the window was larger before senior age, also with overlapping intervals: 0.15 s on the 100 m at 12–13 against 0.11 s at 20–34, and 0.25 against 0.14 s on the 200 m. As a share of race time it was smaller for boys aged 12–13 than for senior men on the 100 m (1.10 against 1.27 % of the mean time) and only slightly larger for girls than for senior women (0.95 against

0.87 %). Relative to the athlete's own variability the window is worth 0.29–1.26 of the robust scale across events, sexes and ages, and most among senior sprinters, whose marks scatter least (men's 100 m: 0.67 at 12–13 against 1.26 at 20–34). Two external yardsticks complete the picture. On the federation's age-graded scoring tables, +2.0 m/s on the 100 m converts to 22–29 points at age 13 against 17–25 points on the oldest (age-19) table, a youth margin of about the same proportion as in seconds. And in the 2025 national 100 m season lists (legal marks only, wind $\leq$ +2.0 m/s; senior lists open to all ages), correcting tailwind-aided marks to still air with the class-specific slopes brings one new athlete into the top 10 of the girls' class-14 list and of the senior men's list and two into the senior women's list (none in the boys' class-14 list); in the senior lists 66 % (men) and 56 % (women) of top-50 marks carried tailwinds above +1 m/s: wind moves rankings at youth and senior ages alike.

Table 6: Advantage of a +2.0 m/s tailwind over still air, valued from the slope over 0 to +2 m/s: seconds, and in parentheses the share of the athlete's robust within-season scale of wind-adjusted performance.

Event	Sex	10–11	12–13	14–15	16–17	18–19	20–34
60 m	Boys/men	0.11 s (0.64)	0.13 s (0.90)	0.11 s (0.96)	0.09 s (0.99)	—	—
60 m	Girls/women	0.11 s (0.66)	0.11 s (0.83)	0.09 s (0.76)	0.06 s (0.59)	—	—
100 m	Boys/men	—	0.16 s (0.67)	0.15 s (0.91)	0.14 s (1.03)	0.16 s (1.25)	0.15 s (1.26)
100 m	Girls/women	—	0.15 s (0.68)	0.15 s (0.82)	0.13 s (0.86)	0.14 s (1.04)	0.11 s (0.90)
200 m	Boys/men	0.27 s (0.37)	0.16 s (0.29)	0.16 s (0.41)	0.19 s (0.66)	0.13 s (0.47)	0.14 s (0.53)
200 m	Girls/women	0.51 s (0.62)	0.25 s (0.45)	0.20 s (0.47)	0.18 s (0.51)	0.16 s (0.50)	0.14 s (0.52)

Seconds are 2$\times$ the slope over 0 to +2 m/s from a piecewise model with separate slopes for headwinds and for tailwinds above +2 (Table S3 gives the slopes). The robust scale is 1.4826 $\times$ the median absolute deviation of within-cell residuals of wind-adjusted marks over cells with at least three results, per event $\times$ sex $\times$ band — a denominator largely free of the wind-driven variance itself, of singleton-cell bias and of gross recording errors. Values reproduced from the estimation files available on request. The 60 m rows at 16–17 and the 200 m rows at 10–11 rest on thin samples.

3.7 Robustness

Estimates were essentially unchanged when cells required three or more results, when exact-zero readings were excluded, when readings within $\pm$0.2 of +2.0 were excluded, across period

halves (except the triple jumps, whose 2021–2026 estimates were 14–24 % higher), and under athlete instead of meet clustering (Table S2; point estimates are identical under either clustering). Replacing month dummies with meet fixed effects — which compares results only within the same meet and absorbs any meet-level weather or gauge siting — attenuated the 60, 100 and 200 m estimates by 13–41 % (100 m: 0.082 to 0.066 s per m/s), left the small 80 and 150 m events essentially unchanged, and raised the jump estimates by 26–47 %. Only 25–34 % of the sprint wind variance lies within meets (39–51 % in the jumps, where attempts spread across the day), so the meet-FE estimates rest on far less variation. The sprint attenuation is compatible with gauge error that weighs more once between-meet variation is removed, but also with some between-meet confounding in the month-dummy estimates; these data cannot separate the two, and the rise in the jumps shows that meet-level factors can push either way. The age gradient survived meet fixed effects: in every sprint series the estimate fell from the youngest to the oldest well-measured band (men's 100 m: 0.107 s per m/s at 12–13 to 0.060 at 20–34; Table S6).

4 Discussion

4.1 Principal findings

In the first population-scale estimates of wind effects across age and ability, a metre per second of wind (linear fit over head- and tailwinds) was worth 42–70 % more, in absolute terms, to a boy sprinter than to a senior man — 0.10 versus 0.07 s on 100 m at 12–13, 0.14 versus 0.08 s on 200 m at 10–11 — and 34–47 % more among girls aged 12–13, with the published elite values (Linthorne, 1994; Mureika, 2001; Moinat et al., 2018; Hollings et al., 2012), and the 0.048–0.059 s per m/s that the same estimator returns in collegiate and world-class 100 m samples (Table S8), emerging as the fast endpoint of a continuous gradient. On the tailwind side alone the boys' 100 m contrast was small (0.071 versus 0.064 s, overlapping intervals) and the girls' larger (0.083 versus 0.057 s). The gradient's dominant mechanism is exposure time: within the registry, as within the collegiate sample, effects scaled approximately with race duration, across age and equally across ability within age, and the percentage effect was accordingly

much flatter — though still declining. Across samples the proportionality is only approximate: per second of race time the collegiate and world-class 100 m slopes lie 15–27 % below the registry's, and at equal race times the registry's seniors gain a fifth to a quarter more than the slowest collegiate tertile, so no single per-second rate transfers between registries unchanged. The residual decline cannot be the aerodynamic share, which grows with speed and would push the percentage effect the other way; it is consistent with a fixed wind being a larger fraction of a slower runner's airspeed, and perhaps with children's higher surface-to-mass ratio. Headwinds cost sprinters more than tailwinds gave them, by about as much as drag physics predicts for the observed winds on the 100 m, by less on the 60 m, and by more on the 200 m, where the gauge covers only the straight; in a post-plan extension, class-specific short hurdles matched flat sprints over 60 m and mostly exceeded them over 100 and 110 m, partly through longer race times.

4.2 Asymmetry, speed and the psychological interpretation

An asymmetry larger than an elite benchmark does not by itself point to a psychological effect. Lybbert and Zheng (2023) found that headwinds slowed US high-school sprinters more than a drag model for elite men predicts — head/tail ratios of 1.74 for boys and 2.08 for girls in their athlete fixed-effects estimates — and attributed the excess, larger for girls, to mindset. But the drag asymmetry itself grows as speed falls, roughly as $(2v + w)/(2v - w)$: over our observed 100 m winds, drag-only physics passed through the same estimator gives a ratio of 1.43 at a uniform elite 10.0 s, 1.57 at their boys' mean of 12.60 s and 1.69 at their girls' mean of 14.56 s, and more in samples of mixed speed, where the slowest runners carry the largest ratios (1.67 for our boys and men, averaging 12.08 s, and 1.73 for our girls and women, averaging 13.53 s). Where the benchmark is matched to the estimator and to each athlete's own speed, the 100 m asymmetry equals the drag prediction for girls and women and falls short of it for boys and men; the sex difference follows the speed difference, and the asymmetry is, if anything, smaller within meets (Table S7). Their boys' ratio is thus about what drag predicts for runners of their speed. Their girls' ratio remains above the 1.73 of our faster girls and women, but an unreported wind distribution and the absence of meet-level controls — headwinds often come with cold,

wet conditions — leave physics and weather as candidate explanations for that remainder. Their further argument, that this psychological slack let a viral record shift subsequent times, rests on this premise. The external samples sharpen the point: collegiate sprinters, between the high-school and the elite samples in speed, match drag at their own speeds as the registry does, and only the championship finalists exceed it (1.9 against 1.45–1.49) — on the tailwind side, since a tailwind is worth least to them (0.036 and 0.044 s per m/s) while their headwind slopes (0.070 and 0.083) are about the collegiate ones (0.077 and 0.081). Selection on favourable readings at the fast end, not a headwind penalty among the slow, is then the natural suspect before mindset.

4.3 Implications

For measurement, the results supply what has been missing: empirically grounded, age- and event-specific wind values for youth athletics, with a full correction table by event, sex and age band in Table S3. Since neither the absolute nor the percentage effect is strictly invariant, corrections should be age-graded; a single senior-derived factor understates youth wind effects by up to two-fifths. For fairness, the ± 2.0 m/s window does not favour the youngest uniformly. In seconds, a legal tailwind is worth about as much to boys aged 12–13 as to senior men and more to girls than to senior women; relative to athletes' own within-season variability it is worth less for the young than for seniors (on the 100 m, 0.67–0.68 of the robust scale at 12–13 against 0.90–1.26 at 20–34). What it does in every age band is move 100 and 200 m marks by more than a tenth of a second, and raw marks are what season lists, records and qualifying standards compare. The 2025 list exercise makes it concrete: still-air correction changes who is in the top 10, senior lists included. The practical conclusions are modest and do not require rule changes: report and store wind with youth results (as Norway does), prefer wind-adjusted or wind-stratified comparisons when marks inform selection or records, and treat single wind-assisted breakthroughs with proportionate scepticism. For talent evaluation the point compounds a broader lesson of the identification literature — early performance is an unreliable selection signal (Vaeyens et al., 2008; Guttormsen, 2026): at the ages where selection decisions lean hardest on raw times, a metre per second of wind moves those times by more seconds than at senior level.

4.4 Strengths and limitations

Strengths include scale (392 797 analysed results, 295 293 of them identifying; 74 110 athlete-season-event cells), a within-athlete design that removes stable athlete and seasonal composition, direction-and-magnitude agreement with the elite literature at the senior endpoint and, with the same estimator, with collegiate and world-class samples (Table S8), internal consistency across age, ability, events and the hurdles extension, and a plan committed before estimation. Limitations are stated as read rules: gauge readings mismeasure athlete-experienced wind, so to the extent the error is classical the sprint estimates are lower bounds, and the meet-FE attenuation shows how much identification leans on between-meet wind variation; the gauge protocol differs by event family, and for the bend events measures only part of the course, so cross-family comparisons (and the low per-second rates of the 150/200 m) partly reflect measurement rather than physiology; wind is not randomised, and within-season form-weather covariation is controlled only by month, round and meet terms; jump estimates use the wind of the measured attempt only, so their bias direction is not established; the event-wide plausibility bounds admit a few gross recording errors, which barely move the slopes but motivated the robust scale in H5; the senior stratum is 20–34, so masters athletics is outside scope; missing wind (4.8 % of approved outdoor results, concentrated in the smallest meets) limits completeness but gives no mechanism for bias within athlete-seasons; temperature and other weather are unobserved (a meteorological linkage is a natural extension); and the setting is one federation's registry: the mechanisms are physical and the external samples reproduce the pattern, but not the per-second rate (Table S8).

5 Conclusion

Wind effects in athletics are not a constant to be borrowed from elite studies: the same estimator gives 0.048–0.059 s per m/s on the 100 m in collegiate and world-class samples against 0.082 in the registry. Per m/s of wind they are 34–70 % larger for child sprinters than for seniors, largely because slower athletes are exposed longer; they fall with ability for the same reason; and headwinds cost more than tailwinds give, by about as much as drag physics predicts on

the 100 m. Within the legal window a +2.0 m/s tailwind is worth 0.11–0.25 s on the 100 and 200 m from age 12 — roughly the same across ages for boys and men, more for girls than for senior women — enough to reorder season lists at youth and senior ages alike. The remedy requires no new rules, because it operates on lists rather than races: wind is already recorded with every result, and the age-specific corrections estimated here (Table S3) can make lists, records and talent evaluation measurably fairer — for seniors too — while every race is run exactly as before.

Declarations

Ethics. Competition results registered by the Norwegian Athletics Federation (publicly listed from age 13; registered but not publicly released for the 10–12 age classes); no individual is identified; performance data below age 13 are used only to estimate measurement conditions, never to rank or evaluate children. Ethics committee approval was not required: the study is not medical or health research as defined by Norway's Health Research Act (Helseforskningsloven §§ 2 and 4), and Sikt (the Norwegian Agency for Shared Services in Education and Research) confirmed that the use of the data raises no data-protection concerns. No participants were contacted or recruited, and informed consent was not required. The study was conducted in accordance with the Declaration of Helsinki.

Data availability. The competition records used in this study are publicly accessible via the Norwegian Athletics Federation's online register for ages 13 and above; records for the 10–12 age classes are not publicly released by the federation. The derived analysis dataset cannot be publicly shared because it links the records to athletes' birth years and therefore contains personal data under the EU General Data Protection Regulation (GDPR). The full analysis code, with the aggregate estimation files, will be deposited in a public repository upon acceptance. The external samples (Table S8) are drawn from the public athlete profiles of the Track & Field Results Reporting System (TFRRS; https://www.tfrrs.org) and from Tilastopaja (Tilastopaja Oy, Turku, Finland), a subscription database whose athlete-level records are not redistributed.

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Supplementary material

Table S1: Wind effect in the horizontal jumps by age band (gain per +1 m/s; values girls/women / boys/men; 95 % CIs clustered on meet).

Event	Unit	10–11	12–13	14–15	16–17	18–19	20–34
Long jump	cm	—	—	2.89 (2.42–3.35) / 3.36 (2.82–3.89)	3.04 (2.48–3.60) / 3.71 (2.68–4.74)	3.20 (2.40–3.99) / 2.94 (1.58–4.29)	4.15 (3.03–5.27) / 3.11 (1.69–4.52)
Long jump	%	—	—	0.70 (0.58–0.82) / 0.76 (0.62–0.90)	0.62 (0.49–0.76) / 0.66 (0.44–0.89)	0.64 (0.46–0.82) / 0.48 (0.23–0.73)	0.76 (0.53–1.00) / 0.49 (0.23–0.75)
Long jump (zone)	cm	1.83 (1.41–2.24) / 2.39 (1.88–2.90)	1.86 (1.59–2.14) / 2.54 (2.17–2.91)	—	—	—	—
Long jump (zone)	%	0.55 (0.42–0.68) / 0.70 (0.54–0.85)	0.49 (0.41–0.56) / 0.66 (0.56–0.75)	—	—	—	—
Triple jump	cm	—	—	4.18 (3.07–5.29) / 4.20 (2.49–5.91)	4.13 (2.88–5.37) / 4.10 (2.09–6.11)	3.52 (1.68–5.36) / 4.21 (1.67–6.76)	4.43 (3.14–5.72) / 4.89 (2.55–7.23)
Triple jump	%	—	—	0.45 (0.33–0.57) / 0.41 (0.24–0.58)	0.40 (0.28–0.52) / 0.32 (0.15–0.49)	0.31 (0.13–0.49) / 0.32 (0.12–0.52)	0.37 (0.26–0.48) / 0.34 (0.17–0.51)

In each cell the first value is girls/women, the second boys/men. The zone events are contested only in the youngest classes. In centimetres the effect rises with age for girls and women as jump distances and flight times grow; in percent it shows no consistent age direction.

Table S2: Robustness: gain per +1 m/s under alternative specifications (point estimates; s for sprints, cm for jumps).

Event	Primary	Cells $\geq$3	No exact 0.0	Excl. $\pm$0.2 of +2.0	2011–19	2021–26	Meet FE
60 m	0.0591	0.0590	0.0591	0.0587	0.0600	0.0573	0.0513
80 m	0.0746	0.0711	0.0761	0.0729	0.0740	0.0779	0.0768
100 m	0.0823	0.0816	0.0823	0.0829	0.0843	0.0795	0.0659
150 m	0.0814	—	0.0780	0.0850	—	—	0.0838
200 m	0.1215	0.1203	0.1215	0.1212	0.1258	0.1135	0.0713
Long jump	3.1597	3.1129	3.2467	3.1932	3.2987	2.9343	4.1017
Long jump (zone)	2.1133	2.0471	2.1538	2.0984	2.1337	2.1518	2.6624
Triple jump	4.1848	4.1647	4.2642	4.1691	4.0189	4.7732	6.1704
Triple jump (zone)	3.8821	3.3154	3.8589	3.7725	3.7075	4.8247	5.0930

Point estimates are identical under athlete and meet clustering; only intervals differ. The 150 m lacks the cells-$\geq$3 and period columns for want of data.

Table S3: Correction table: tailwind slope over 0 to +2 m/s, and separate tail- and headwind slopes, by event, sex and age band (sprints; jumps event-level, in cm), with 95 % CIs clustered on meet.

Event	Sex	Age band	Tailwind 0 to +2	Tailwind slope per +1 m/s (95 % CI)	Headwind slope per −1 m/s (95 % CI)	n
60 m	Boys/men	10–11	0.057	0.055 (0.046–0.064)	0.088 (0.077–0.098)	7 635
60 m	Boys/men	12–13	0.066	0.056 (0.050–0.062)	0.071 (0.064–0.078)	15 457
60 m	Boys/men	14–15	0.054	0.042 (0.036–0.049)	0.062 (0.051–0.073)	6 955
60 m	Boys/men	16–17	0.046	0.019 (0.004–0.035)	0.064 (0.049–0.080)	570
60 m	Girls/women	10–11	0.055	0.044 (0.036–0.052)	0.085 (0.076–0.094)	9 794
60 m	Girls/women	12–13	0.057	0.047 (0.042–0.051)	0.067 (0.062–0.072)	21 886
60 m	Girls/women	14–15	0.046	0.038 (0.032–0.043)	0.068 (0.062–0.075)	9 320
60 m	Girls/women	16–17	0.030	0.021 (0.007–0.034)	0.091 (0.068–0.113)	389
100 m	Boys/men	12–13	0.082	0.071 (0.041–0.102)	0.141 (0.111–0.170)	1 755
100 m	Boys/men	14–15	0.073	0.072 (0.063–0.081)	0.113 (0.102–0.123)	7 499
100 m	Boys/men	16–17	0.069	0.062 (0.054–0.070)	0.102 (0.094–0.111)	9 879
100 m	Boys/men	18–19	0.079	0.067 (0.058–0.077)	0.093 (0.080–0.107)	6 030
100 m	Boys/men	20–34	0.073	0.064 (0.058–0.071)	0.082 (0.071–0.092)	8 909
100 m	Girls/women	12–13	0.073	0.083 (0.062–0.105)	0.112 (0.091–0.134)	2 733
100 m	Girls/women	14–15	0.073	0.071 (0.063–0.079)	0.126 (0.116–0.135)	9 740
100 m	Girls/women	16–17	0.066	0.059 (0.052–0.066)	0.112 (0.102–0.122)	9 062
100 m	Girls/women	18–19	0.072	0.064 (0.057–0.072)	0.106 (0.092–0.120)	4 207
100 m	Girls/women	20–34	0.055	0.057 (0.050–0.064)	0.098 (0.087–0.109)	4 543
200 m	Boys/men	10–11	0.133	0.107 (0.023–0.191)	0.194 (0.090–0.298)	1 301
200 m	Boys/men	12–13	0.081	0.076 (0.041–0.111)	0.217 (0.174–0.259)	5 443
200 m	Boys/men	14–15	0.079	0.078 (0.059–0.096)	0.188 (0.164–0.212)	7 147
200 m	Boys/men	16–17	0.094	0.074 (0.060–0.088)	0.172 (0.146–0.198)	6 372
200 m	Boys/men	18–19	0.063	0.069 (0.053–0.084)	0.149 (0.124–0.173)	4 158
200 m	Boys/men	20–34	0.070	0.048 (0.026–0.070)	0.151 (0.125–0.178)	4 931
200 m	Girls/women	10–11	0.256	0.099 (0.025–0.173)	0.086 (−0.012–0.185)	1 780
200 m	Girls/women	12–13	0.127	0.105 (0.080–0.130)	0.181 (0.143–0.219)	7 710
200 m	Girls/women	14–15	0.100	0.094 (0.075–0.113)	0.186 (0.162–0.211)	8 640
200 m	Girls/women	16–17	0.091	0.066 (0.049–0.082)	0.208 (0.173–0.243)	5 854
200 m	Girls/women	18–19	0.081	0.065 (0.047–0.082)	0.151 (0.120–0.183)	2 956
200 m	Girls/women	20–34	0.071	0.047 (0.025–0.070)	0.180 (0.143–0.218)	2 954
Long jump	Both	all	—	2.93 (2.58–3.28) cm	3.71 (2.77–4.65) cm	35 774
Long jump (zone)	Both	all	—	2.48 (2.19–2.77) cm	1.41 (0.95–1.87) cm	44 336
Triple jump	Both	all	—	4.63 (3.78–5.48) cm	2.79 (0.78–4.80) cm	11 658
Triple jump (zone)	Both	all	—	4.69 (3.40–5.98) cm	1.90 (−0.25–4.04) cm	5 426

Worked example: a girls' 12–13 100 m run in +1.8 m/s is corrected by $1.8 \times 0.073 = 0.13$ s to still air, using the slope over 0 to +2 m/s. Sprint slopes are in seconds; the four jump rows are in centimetres, pooled across bands. The 60 m rows at 16–17 rest on thin outdoor samples.

Table S4: Effect per second of race time: gain per +1 m/s divided by mean race duration, flat sprints and short hurdles.

Event	Mean time (s)	Gain per +1 m/s (s)	Per second of race time
60 m	9.67	0.059	0.00611
80 m	12.58	0.075	0.00594
100 m	13.09	0.082	0.00629
150 m	20.22	0.081	0.00403
200 m	28.47	0.121	0.00427
60 m hurdles 68 cm	13.16	0.063	0.00478
60 m hurdles 76.2 cm	11.82	0.062	0.00525
80 m hurdles 76.2 cm	13.36	0.072	0.00535
80 m hurdles 84 cm	14.09	0.078	0.00551
100 m hurdles 76.2 cm	15.61	0.108	0.00694
100 m hurdles 84 cm	14.94	0.078	0.00523
100 m hurdles 91.4 cm	15.66	0.084	0.00534
110 m hurdles 91.4 cm	15.90	0.103	0.00646
110 m hurdles 100 cm	15.74	0.096	0.00608
110 m hurdles 106.7 cm	15.53	0.084	0.00540

Straight flat sprints cluster at 0.0059–0.0063 per m/s per second of race time; the bend events (150/200 m) sit about a third lower; hurdle codes bracket the flat rates.

Table S5: Quadratic specification: linear and quadratic wind terms (raw outcome units per m/s: seconds for sprints, metres for jumps, e.g. 0.0328 m = 3.28 cm; SEs clustered on meet).

Event	Linear term (SE)	Quadratic term (SE)	n
60 m	−0.0596 (0.0010)	0.0037 (0.0004)	72 558
80 m	−0.0752 (0.0065)	0.0013 (0.0026)	1 410
100 m	−0.0856 (0.0014)	0.0057 (0.0005)	64 557
150 m	−0.0794 (0.0194)	−0.0052 (0.0109)	328
200 m	−0.1307 (0.0038)	0.0141 (0.0013)	59 246
Long jump	0.0328 (0.0023)	−0.0008 (0.0008)	35 774
Long jump (zone)	0.0194 (0.0012)	0.0015 (0.0004)	44 336
Triple jump	0.0382 (0.0047)	0.0020 (0.0016)	11 658
Triple jump (zone)	0.0342 (0.0060)	0.0030 (0.0018)	5 426

Negative linear terms for sprints mean faster times with tailwind. Positive quadratic terms in the large sprints match the headwind-tailwind asymmetry of Table 4.

Table S6: Meet fixed effects by age band: effect per m/s of wind with month dummies replaced by meet fixed effects (seconds, 95 % CI clustered on meet), sprints.

Event	Sex	10–11	12–13	14–15	16–17	18–19	20–34
60 m	Boys/men	0.055 (0.045–0.066)	0.055 (0.049–0.061)	0.047 (0.040–0.055)	0.030 (0.011–0.048)	—	—
60 m	Girls/women	0.056 (0.049–0.063)	0.049 (0.043–0.054)	0.047 (0.041–0.052)	0.030 (0.012–0.048)	—	—
100 m	Boys/men	—	0.107 (0.072–0.143)	0.071 (0.056–0.085)	0.063 (0.050–0.075)	0.060 (0.046–0.073)	0.060 (0.051–0.069)
100 m	Girls/women	—	0.093 (0.077–0.109)	0.077 (0.066–0.089)	0.065 (0.057–0.072)	0.069 (0.059–0.080)	0.062 (0.054–0.070)
200 m	Boys/men	−0.000 (−0.147–0.146)	0.091 (0.054–0.128)	0.078 (0.054–0.102)	0.063 (0.038–0.088)	0.058 (0.026–0.090)	0.053 (0.033–0.072)
200 m	Girls/women	0.030 (−0.082–0.142)	0.087 (0.060–0.113)	0.080 (0.061–0.100)	0.079 (0.057–0.100)	0.071 (0.047–0.094)	0.059 (0.006–0.111)

Same reporting threshold as Table 3. The 200 m at 10–11 has too few results per meet for the meet-FE estimate to be informative.

Table S7: Headwind/tailwind asymmetry on the 100 m by sex and age band, against drag-only physics at the athletes' own speeds (slopes in seconds per m/s, signed as gains with tailwind; 95 % CIs clustered on meet).

Sex	Age band	n	Mean time (s)	Tailwind slope	Headwind slope	Head/tail ratio (95 % CI)	Drag-only ratio
Girls/women	12–13	2 733	14.80	0.083	0.112	1.35 (0.82–1.88)	1.81
Girls/women	14–15	9 740	13.93	0.071	0.126	1.76 (1.48–2.04)	1.73
Girls/women	16–17	9 062	13.41	0.059	0.112	1.91 (1.59–2.22)	1.68
Girls/women	18–19	4 207	13.04	0.064	0.106	1.66 (1.31–2.01)	1.66
Girls/women	20–34	4 543	12.57	0.057	0.098	1.71 (1.39–2.03)	1.58
Girls/women	All ages	30 392	13.53	0.065	0.114	1.77 (1.58–1.96)	1.73
Girls/women	All ages, meet FE	30 392	13.53	0.058	0.092	1.59 (1.35–1.83)	1.70
Boys/men	12–13	1 755	14.42	0.071	0.141	1.97 (0.87–3.07)	1.92
Boys/men	14–15	7 499	12.79	0.072	0.113	1.56 (1.28–1.85)	1.73
Boys/men	16–17	9 879	12.01	0.062	0.102	1.65 (1.37–1.93)	1.59
Boys/men	18–19	6 030	11.61	0.067	0.093	1.39 (1.03–1.75)	1.55
Boys/men	20–34	8 909	11.36	0.064	0.082	1.27 (1.01–1.52)	1.58
Boys/men	All ages	34 165	12.08	0.066	0.101	1.54 (1.36–1.71)	1.67
Boys/men	All ages, meet FE	34 165	12.08	0.055	0.080	1.46 (1.18–1.74)	1.63

Head/tail ratio: headwind slope over tailwind slope, CI by the delta method. Drag-only ratio: time changes proportional to the change in squared airspeed at each athlete-season's mean speed, imposed on the observed winds and estimated with the same model (with meet fixed effects in the meet-FE rows). Added after the analysis plan.

Table S8: The elite end with the same estimator: wind effects in US collegiate results (TFRRS) and in the career results of championship finalists (Tilastopaja), with the registry's all-age and senior rows as reference (within athlete × season; 95 % CIs and SEs clustered on meet).

Sample	n	Mean mark	Per m/s, linear (95 % CI)	Per s	Tailwind	Headwind	Head/tail (drag)	0 to +2	Above +2 (SE)	Meet FE
100 m, men										
Registry, all ages	34 165	12.40	0.080 (0.077–0.083)	0.0064	0.066	0.101	1.54 (1.67)	0.073	0.049 (0.005)	0.064
Registry, 20–34	8 909	11.42	0.071 (0.066–0.075)	0.0062	0.064	0.082	1.27 (1.58)	0.073	0.047 (0.007)	0.060
US collegiate (TFRRS)	57 534	10.89	0.054 (0.053–0.056)	0.0050	0.047	0.077	1.62 (1.67)	0.051	0.043 (0.002)	0.045
Championship finalists (Tilastopaja)	8 885	10.19	0.048 (0.045–0.051)	0.0047	0.036	0.070	1.93 (1.45)	0.038	0.033 (0.005)	0.040
100 m, women										
Registry, all ages	30 392	13.84	0.085 (0.082–0.089)	0.0062	0.065	0.114	1.77 (1.73)	0.067	0.059 (0.006)	0.071
Registry, 20–34	4 543	12.63	0.072 (0.067–0.078)	0.0057	0.057	0.098	1.71 (1.58)	0.055	0.063 (0.008)	0.062
US collegiate (TFRRS)	63 234	12.22	0.058 (0.056–0.060)	0.0047	0.050	0.081	1.61 (1.76)	0.059	0.041 (0.002)	0.047
Championship finalists (Tilastopaja)	8 047	11.25	0.059 (0.055–0.063)	0.0052	0.044	0.083	1.86 (1.49)	0.047	0.039 (0.008)	0.053
200 m, men										
Registry, all ages	29 352	27.07	0.117 (0.108–0.127)	0.0043	0.071	0.183	2.56 (1.82)	0.081	0.053 (0.015)	0.067
Registry, 20–34	4 931	23.08	0.085 (0.068–0.101)	0.0037	0.048	0.151	3.15 (1.64)	0.070	0.013 (0.027)	0.053
US collegiate (TFRRS)	57 111	22.05	0.066 (0.062–0.069)	0.0030	0.052	0.104	2.00 (1.64)	0.057	0.045 (0.005)	0.050
Championship finalists (Tilastopaja)	6 265	20.54	0.065 (0.057–0.074)	0.0032	0.030	0.114	3.74 (1.41)	0.034	0.021 (0.016)	0.068
200 m, women										
Registry, all ages	29 894	29.86	0.125 (0.115–0.136)	0.0042	0.083	0.185	2.22 (1.87)	0.110	0.033 (0.015)	0.077
Registry, 20–34	2 954	25.83	0.095 (0.075–0.115)	0.0037	0.047	0.180	3.81 (1.76)	0.071	0.011 (0.029)	0.059
US collegiate (TFRRS)	71 467	25.30	0.074 (0.070–0.078)	0.0029	0.057	0.118	2.06 (1.81)	0.072	0.040 (0.005)	0.056
Championship finalists (Tilastopaja)	5 804	22.93	0.078 (0.069–0.088)	0.0034	0.055	0.110	2.00 (1.50)	0.063	0.037 (0.022)	0.079
100 m hurdles, women										
US collegiate (TFRRS)	61 051	14.74	0.075 (0.072–0.079)	0.0051	0.057	0.129	2.27	0.072	0.038 (0.005)	0.062
Championship finalists (Tilastopaja)	7 302	12.94	0.054 (0.049–0.059)	0.0041	0.033	0.084	2.51	0.031	0.039 (0.010)	0.045
110 m hurdles, men										
US collegiate (TFRRS)	45 225	14.97	0.057 (0.054–0.060)	0.0038	0.043	0.099	2.31	0.051	0.034 (0.004)	0.043
Championship finalists (Tilastopaja)	6 788	13.47	0.044 (0.039–0.050)	0.0033	0.026	0.072	2.77	0.027	0.023 (0.009)	0.046
Long jump, men										
Registry, all ages	15 259	4.69	3.34 (2.90–3.78)	—	2.98	4.27	1.43	1.54	5.73 (0.60)	4.77
Registry, 20–34	1 813	6.26	3.11 (1.69–4.52)	—	2.85	4.17	1.46	2.70	3.11 (1.71)	8.56
US collegiate (TFRRS)	38 335	6.77	3.74 (3.46–4.02)	—	3.88	3.18	0.82	4.97	2.58 (0.31)	3.35
Championship finalists (Tilastopaja)	6 279	7.95	4.53 (3.95–5.11)	—	4.08	6.42	1.57	4.89	2.40 (0.94)	4.13
Long jump, women										
Registry, all ages	20 515	4.17	3.04 (2.70–3.38)	—	2.88	3.40	1.18	2.51	3.58 (0.43)	4.18
Registry, 20–34	1 536	5.43	4.15 (3.03–5.27)	—	3.99	4.88	1.22	4.38	3.42 (1.26)	7.20
US collegiate (TFRRS)	50 068	5.44	2.73 (2.52–2.94)	—	2.86	2.25	0.79	3.81	1.65 (0.23)	2.37
Championship finalists (Tilastopaja)	6 290	6.58	3.71 (3.27–4.16)	—	3.41	4.96	1.46	4.62	1.04 (0.58)	2.84
Triple jump, men										
Registry, all ages	4 237	11.15	4.21 (3.14–5.28)	—	5.18	1.27	0.25	3.39	8.50 (1.60)	7.39
Registry, 20–34	527	13.66	4.89 (2.55–7.23)	—	7.68	−5.28	-0.69	3.94	14.51 (4.48)	10.77
US collegiate (TFRRS)	24 224	14.25	4.57 (4.15–5.00)	—	4.27	5.76	1.35	5.54	2.68 (0.57)	3.56
Championship finalists (Tilastopaja)	4 801	16.85	6.55 (5.45–7.65)	—	5.74	9.50	1.65	5.83	5.54 (1.46)	5.08
Triple jump, women										
Registry, all ages	7 421	9.89	4.15 (3.41–4.90)	—	4.30	3.70	0.86	3.79	5.28 (1.14)	6.02
Registry, 20–34	1 018	11.71	4.43 (3.14–5.72)	—	5.16	0.49	0.10	5.91	3.86 (1.80)	6.30
US collegiate (TFRRS)	32 378	11.68	3.94 (3.64–4.25)	—	3.78	4.60	1.22	4.90	2.34 (0.39)	2.88
Championship finalists (Tilastopaja)	4 467	14.21	5.89 (5.09–6.68)	—	5.86	6.00	1.02	6.88	3.66 (1.22)	5.28

Sprints in seconds, jumps in centimetres, per m/s of wind, signed as gains with tailwind; n is the number of results in athlete-season-event cells with at least two wind-measured marks. Mean mark: all analysed results (s for sprints, m for jumps), as in Table S4. Per s: the linear gain divided by the mean time. The registry's 20–34 jump rows rest on thin samples. Tailwind and headwind: the separate branch slopes; head/tail: their ratio, with in parentheses the ratio drag-only physics predicts for the same winds and estimator at the athletes' own speeds (100 m and 200 m). 0 to +2 m/s and above +2 m/s: the tailwind slopes of the piecewise model used for H5. Meet FE: the linear slope with month dummies replaced by meet fixed effects. The registry's hurdle events are class-specific (Table 5), so no registry row is given for the hurdles. The external estimates were